

AMAR FAUZIE

Data Scientist | Machine Learning Engineer | AI Automation Engineer

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PROFESSIONAL SUMMARY

AI/ML practitioner skilled in Python, RAG pipelines, semantic search, vector databases, and AI agent automation. Builds end-to-end systems from data preprocessing through model evaluation and backend deployment, with hands-on Arabic NLP project experience as a differentiator.

SKILLS

Python, SQL, Pandas/NumPy/Scikit-learn, TensorFlow, XGBoost, RAG, LLMs, Semantic Search, Text Embeddings, LangChain, CrewAI, Qdrant/Weaviate, FastAPI, SQLAlchemy, Pydantic, n8n Automation

KEY PROJECTS

Diwanic — Arabic Poetry Semantic Retrieval System

- Built a semantic retrieval engine for Arabic poetry using multilingual embeddings (BAAI/bge-m3) with hybrid vector + keyword search and Reciprocal Rank Fusion. Backend via FastAPI, PostgreSQL, and Qdrant; corpus ingestion orchestrated with Prefect; Gradio and React interfaces.

Master Advisor Agent — LangGraph Multi-Agent Pipeline

- Designed a LangGraph state-machine pipeline that automates the search for a fully funded Master's program: analyze profile, search real universities and scholarships via Tavily, rank matches by fit score, then an LLM judge approves or rejects with a feedback retry loop until results pass. OpenAI GPT-4o-mini, SQLAlchemy with SQLite persistence, structured logging.

Hadith Research Agent — Multi-Agent Verification Pipeline

- Built a multi-agent verification pipeline for Islamic texts where hallucination is unacceptable: CrewAI agents search trusted sources (sunnah.com, Maktabah Shamela), cross-validate narrations, and every claim links back to its source.

Fraud / Phishing Detector — Supervised ML Classification

- End-to-end ML project from an AI bootcamp hackathon: sourced and cleaned a Kaggle dataset, trained scikit-learn classification models, and deployed the app on AWS. Completed solo after the team stopped; covered the full workflow from data to deployment.

HadithGraph

- In development: An AI-powered hadith research system that combines semantic search with graph-based retrieval to improve answer accuracy and uncover relationships across the hadith corpus. It uses graph connections to support automated takhrij by tracing related narrations, sources, and chains of transmission

WORK EXPERIENCE

Hajj Pilgrimage Guide / Mutawwif | Tabung Haji

2024 – 2026

- Communicated religious, logistical, and safety information clearly to pilgrims from diverse backgrounds; managed coordination and problem-solving in a high-pressure, time-sensitive environment.

Company Representative / Exhibitor | Siiru — Hajj & Umrah Forum, King Salman Int'l Convention Centre

- Presented company offerings and communicated with diverse audiences in a professional, public-facing exhibition environment.

Committee Member (AJK) | Nadi Pencinta Madinah

- Co-developed educational modules and organized cultural activities and historical tours across Makkah, Madinah, and Taif for Malaysian students.

President | ARCOM, International Islamic University Malaysia

2019-2020

- Led a student organization focused on Arabic communication; organized competitions and represented the group in Arabic debate and public speaking.

President | Sejahtera, Universiti Sultan Zainal Abidin

2018

- Led organizational activities and built the organization's social media and digital communication presence.

CERTIFICATIONS

- DeepLearning.AI: Multi AI Agent Systems with CrewAI, Retrieval Augmented Generation (RAG), Pydantic for LLM Workflows
- DeepLearning.AI: Probability & Statistics for ML & Data Science, Advanced Learning Algorithms, Linear Algebra for ML
- DeepLearning.AI: Supervised Machine Learning — Regression and Classification
- Nexperts Academy: AI & Machine Learning Bootcamp (sponsored by Yayasan Peneraju)

EDUCATION

B.A., Islamic Creed and Da'wah | Islamic University of Madinah, Saudi Arabia

CGPA: 4.61/5.00

Diploma, Arabic Language and Literature | Universiti Sultan Zainal Abidin, Malaysia

CGPA: 3.80/4.00

Undergraduate Study, Arabic for International Communication | International Islamic University Malaysia

ADDITIONAL SKILLS

- Languages: Arabic (advanced/professional), Malay (native), English (professional working proficiency).
- Public Speaking & Evaluation: Best Debater (USLUB 2.0, 2019); international Arabic debate representative (IIUM, 2018); debate adjudicator, Terengganu State Arabic Debate Competition (2018).